

Prithvi Chakravarthy

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EDUCATION

DePaul University

Master of Science in Computer Science

Chicago, IL

Sept. 2025 – Present

- Coursework: Distributed Systems, Applied Algorithms and Structures, DBMS, Real-Time System Architecture

BITS Pilani, WILP

Postgraduate Certificate in AI/ML

India

Oct. 2023 – Oct. 2024

- Coursework: Deep Learning, Neural Networks, Text Mining, Feature Engineering, Unsupervised Learning

VIT Bhopal

Bachelor of Technology in Computer Science

India

Jun. 2018 – Jun. 2022

- Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Architecture, Soft Computing, Computer Networks, Theory of Computation & Compiler Design

EXPERIENCE

Designare Solutions Pvt Ltd

Full Stack Developer

Mumbai, India

March 2022 – August 2025

- Developed and deployed full stack applications using React, Node.js, Python (FastAPI) and PostgreSQL, improving scalability and backend performance.
- Designed distributed REST APIs and optimized data pipelines, reducing query retrieval time by 25%.
- Managed production deployments on Google Cloud infrastructure, maintaining 99.9% uptime.
- Implemented secure user authentication, role-based access control (RBAC), and wired external data sources into cohesive end-user interfaces.

PROJECTS

AI Journal OS | Rust, Tauri, React, TypeScript, SQLite, Candle

- Built a privacy-first, local-offline AI ecosystem using a dual-topology architecture, integrating a Tauri React frontend with a standalone Rust background worker.
- Engineered a local Retrieval-Augmented Generation (RAG) pipeline utilizing pure Rust vector math and SQLite BLOB structures, generating CPU-friendly embeddings with the all-MiniLM-L6-v2 model.
- Designed a privacy-safe AI orchestration pipeline that budgets token counts, redacts PII locally via Hugging Face models, and routes sanitized context to the Groq API before rehydrating tokens upon response.
- Implemented a zero-knowledge cloud synchronization engine with Neon PostgreSQL, ensuring all third-party credentials and personal data remain AES-256 encrypted at rest.

Space Invaders Game Engine | C#, Azul Engine, IrrKlang

- Engineered a high-performance Space Invaders clone using **C#** and the **Azul engine**, strictly adhering to **SOLID principles** and implementing **14+ GoF design patterns** to ensure a modular architecture.
- Optimized memory management by architecting a universal **Object Pool** system (Template pattern), eliminating runtime heap allocation spikes and leveraging **Proxy/Flyweight patterns**.
- Developed a decoupled collision detection system using **Visitor and Observer patterns** for double-dispatch resolution, and managed interval-based events via a **Command pattern** and Priority Queue.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Rust, Kotlin, C#

AI/ML Technologies: Large Language Models (LLMs), RAG Architecture, Prompt Engineering, Embeddings, PyTorch, TensorFlow

Frameworks/Libraries: React, Node.js, FastAPI, Flask, LangChain, LlamaIndex, MaterialUI

Databases & Tools: PostgreSQL, SQLite, Vector Databases, Google Cloud, Docker, REST APIs, Git